

LV Trajectory Engineering Report

Literature-Anchored Reference Profiles: Healthy · HFpEF · DCM/HFrEF

Inline citations for every engineered feature

1. Phase Structure

The cardiac cycle was divided into seven phases at approximately 70 bpm (RR ~850 ms):

Phase	% Cycle	Physiological definition
IVC	0–4%	Isovolumic contraction — pressure rise, zero volume change
Rapid ejection	4–35%	Steep volume reduction, peak twist build-up
Reduced ejection	35–42%	Slower volume loss, peak twist near end-systole
IVR	42–44%	Isovolumic relaxation — rapid pressure fall, zero volume change
Rapid filling	44–57%	E-wave — passive early filling driven by untwist suction
Diastasis	57–85%	Low-flow mid-diastolic plateau
Atrial systole	85–100%	A-wave — atrial kick completes end-diastolic filling

IVR duration compressed (~17 ms vs physiological 60–80 ms) — consistent with mock loop / device setup.

→ Wiggers CJ. The Pressure Pulses in the Cardiovascular System. Longmans, Green, 1928.

→ Kern MJ. The Cardiac Catheterization Handbook, 5th ed., 2011.

→ Atrial kick contribution (~20–30%): StatPearls — "Atrial Kick" (NCBI Bookshelf, 2024).

2. Twist / Rotation

2.1 Healthy

Pre-ejection notch (~–0.8° to –1.4° at t = 0.01–0.02)

A brief clockwise rotation at the apex during IVC before the dominant ejection twist. Reflects early sub-endocardial activation preceding sub-epicardial recruitment.

→ Sengupta PP et al. JACC 2006;47(5):1026–35. doi:10.1016/j.jacc.2005.08.073

→ Sengupta PP et al. JACC Cardiovasc Imaging 2008;1(3):366–76. doi:10.1016/j.jcmg.2008.02.006

→ Twist Mechanics paper, Fig 2 — early negative deflection in apical rotation trace

Peak net twist (~15.5° at t ≈ 0.327)

MRI tagging gives $12.6 \pm 1.5^\circ$ (Lorenz 2000); the Twist Mechanics reference figure shows apical rotation ~16° just before AVC. Engineered peak of ~15.5° sits within the overlap.

→ Lorenz CH, Pastorek JS, Bundy JM. J Cardiovasc Magn Reson 2000;2(2):97–108.

→ Twist Mechanics paper, Fig 2 — apical rotation peak just before phase 3 marker (AVC)

Peak timing (just before AVC, ~35% cycle)

Twist peaks at or just before aortic valve closure across all published modalities.

→ Nakatani S. J Cardiovasc Ultrasound 2011;19(1):1–6. doi:10.4250/jcu.2011.19.1.1

Untwisting onset (begins at/before AVC)

→ Notomi Y et al. Circulation 2006;113:2524–33. doi:10.1161/CIRCULATIONAHA.105.596502

Peak untwisting rate (–115 ± 40 deg/s in IVR)

→ Notomi Y et al. Circulation 2006;113:2524–33.

→ Wang J et al. Circulation 2007;116:2580–6. doi:10.1161/CIRCULATIONAHA.107.706291

IVR untwist fraction (~40–50% of total recoil)

→ Notomi Y et al. Circulation 2006;113:2524–33.

Shoulder/plateau in tail (~3.5° at t = 0.51–0.57)

Visible in apical and net twist traces as clear inflection and plateau before final slow decay.

→ Twist Mechanics paper, Fig 2 — plateau visible in apical rotation trace, phases 3–4

Residual tail (slow decay to 0 by t ≈ 0.90–1.0)

→ Twist Mechanics paper, Fig 2

2.2 Diastolic Dysfunction (HFpEF)

Systolic twist — identical to healthy

Wang and Nagueh reported peak twist $13 \pm 6^\circ$ in diastolic HF vs $14 \pm 5^\circ$ in controls (NS), confirming preserved systolic twist mechanics. All systolic knots were copied exactly from the healthy interpolator.

→ Wang J, Khoury DS, Yue Y, Torre-Amione G, Nagueh SF. Eur Heart J 2008;29(10):1283–9. doi:10.1093/eurheartj/ehn141

IVR untwisting near-absent (10.3° → 9.3°, vs healthy 10.3° → 5.8°)

The defining HFpEF twist signature. Zhang et al. (2020) demonstrated significantly reduced untwisting parameters in HFpEF vs controls; Tan and Sanderson showed delayed untwisting at rest and marked failure to augment on exercise; Takeuchi et al. showed reduction in proportion to LVH severity.

→ Zhang Y et al. Clin Cardiol 2020;43(6):616–22. doi:10.1002/clc.23353

→ Tan YT, Sanderson JE. JACC Cardiovasc Imaging 2009;2(7):802–9.

→ Takeuchi M et al. Eur Heart J 2007;28(19):2313–21. doi:10.1093/eurheartj/ehm340

Bulk of recoil shifted into rapid filling (t = 0.449–0.693)

→ Zhang Y et al. Clin Cardiol 2020;43(6):616–22.

→ Tan YT, Sanderson JE. JACC Cardiovasc Imaging 2009;2(7):802–9.

More diastolic residual (~3° persisting into diastasis)

→ Zhang Y et al. Clin Cardiol 2020;43(6):616–22.

2.3 Systolic Dysfunction (DCM/HFrEF)

Pre-ejection notch near-absent (~–0.2°)

Ventricular dilation in DCM disrupts the oblique fibre architecture that generates the sub-endocardial/sub-epicardial rotational differential.

→ Sengupta PP et al. JACC 2006;47(5):1026–35.

→ Jin SM et al. J Korean Med Sci 2007;22(4):633–8. doi:10.3346/jkms.2007.22.4.633

Peak twist ~4.5–5° (blunted)

→ Wang J et al. Eur Heart J 2008;29(10):1283–9. ($5 \pm 2^\circ$ vs $14 \pm 5^\circ$, $p < 0.001$)

→ Jin SM et al. J Korean Med Sci 2007;22(4):633–8. ($0.3 \pm 2.1^\circ$ severe paediatric DCM)

Delayed peak timing (t ≈ 0.38–0.40, vs healthy 0.327)

→ Jin SM et al. J Korean Med Sci 2007;22(4):633–8.

Slow linear rise — no shoulder feature

Two-component sub-endocardial/sub-epicardial architecture disrupted by fibre reorientation.

→ Lorenz CH et al. J Cardiovasc Magn Reson 2000;2(2):97–108.

→ Sengupta PP et al. JACC 2006;47(5):1026–35.

IVR untwisting near-abolished

→ Wang J et al. Circulation 2007;116:2580–6. (untwisting rate -55 vs -89 deg/s, $p < 0.05$)

→ Jin SM et al. J Korean Med Sci 2007. (apical rotational velocity reversed in DCM)

Long slow featureless tail (proportionally longer vs peak)

→ Jin SM et al. J Korean Med Sci 2007;22(4):633–8.

3. Height / Longitudinal Shortening (MAPSE)

3.1 Healthy

MAPSE ~13 mm (engineered min height ~18 mm)

- CPET-HFpEF cohort (medRxiv 2025): controls MAPSE 12.1 ± 2.2 mm at rest.
- MAPSE >10 mm: 90% sensitivity / 87% specificity for EF >55%. Stevenson D et al. J Emerg Med 2023.

S-curve shortening shape, two-phase (rapid then reduced)

Directly traced from Post-systolic shortening paper, Fig 1 (healthy curve).

- PSS paper, Fig 1

Post-systolic shortening (PSS) — small re-shortening at $t \approx 0.42\text{--}0.44$ after AVC

- PSS paper, Fig 1 — visible as secondary negative dip after AVC marker

Brisk IVR recoil snap

- PSS paper, Fig 1

LV long-axis contribution to stroke volume (~60%)

- Carlsson M et al. Am J Physiol Heart Circ Physiol 2007;292:H1452–9. doi:10.1152/ajpheart.00639.2006

3.2 Diastolic Dysfunction (HFpEF)

Systolic shortening identical to healthy

- Yip GWK et al. Heart 2002;87(5):435–40. doi:10.1136/heart.87.5.435
- Wang J et al. Eur Heart J 2008;29(10):1283–9.

MAPSE mildly reduced ($\sim 10.9 \pm 2.1$ mm vs 12.1 ± 2.2 mm, $p = 0.008$)

- CPET-HFpEF cohort (medRxiv 2025)

Post-systolic shortening more prominent

- PSS paper — HFpEF cohort shows larger PSS amplitude

IVR recoil slower (impaired active relaxation)

- Yip GWK et al. Heart 2002;87(5):435–40.

Early-diastolic recovery rate ~16% reduced

- Codreanu A et al. CMR feature-tracking comparison HFpEF vs controls, 2010.

GLS longitudinal strain ~12% (vs ~19% healthy)

- Wang J et al. Eur Heart J 2008;29(10):1283–9.

3.3 Systolic Dysfunction (DCM/HFrEF)

MAPSE <7 mm (engineered ~8.5 mm)

- Stevenson D et al. J Emerg Med 2023. (MAPSE <7 mm = 92% sensitivity for EF <30%)

Near-linear shortening (no two-phase distinction)

- Carlsson M et al. Am J Physiol Heart Circ Physiol 2007;292:H1452–9.

Delayed onset, flat prolonged trough (elevated MPI)

Reduced dP/dt means slower mechanical activation and prolonged systolic duration.

- Jin SM et al. J Korean Med Sci 2007;22(4):633–8.

PSS absent

Reduced contractility removes the sub-epicardial fibre contribution responsible for PSS.

- PSS paper — absent in low-EF systolic HF cohort

Biphasic recovery with brief plateau (heterogeneous wall motion)

Reflects regional wall motion abnormality — some segments shorten late.

→ CMR wall motion literature; Carlsson M et al. 2007.

Early-diastolic lengthening rate ~50% of healthy

→ Codreanu A et al. CMR feature-tracking 2010.

GLS longitudinal strain ~-4 to -7%

→ Wang J et al. Eur Heart J 2008;29(10):1283–9. (-4% in systolic HF vs -19% controls)

4. Volume

4.1 Healthy

EDV ~120 mL, ESV ~50 mL, EF ~63%

- WASE 3DE Normal Values Study. J Am Soc Echocardiogr 2021.
- Meta-analysis: mean LVEF 62.8% (95% CI 61.0–64.7%). J Cardiovasc Imaging 2025.

Biphasic ejection (steep rapid then shallower reduced)

~2/3 of stroke volume ejected in rapid ejection phase.

- Wiggers CJ. The Pressure Pulses in the Cardiovascular System. 1928.

E-wave dominant ($E/A > 1$), ~55 mL in rapid filling

- Kovacs SJ et al. Am J Physiol Heart Circ Physiol 2004.

A-wave ~15 mL (~20–30% of filling)

- StatPearls — "Atrial Kick" (NCBI Bookshelf, 2024)

4.2 Diastolic Dysfunction (HFpEF)

EF preserved (~57%), ESV slightly higher (~51 mL)

- Wang J et al. Eur Heart J 2008;29(10):1283–9.

E-wave reduced (~28 mL), A-wave enlarged (~37 mL)

Grade I impaired relaxation — reduced early suction from delayed untwisting, compensated by atrial dependence.

- Kovacs SJ et al. Am J Physiol Heart Circ Physiol 2004.
- Notomi Y et al. Circulation 2006;113:2524–33. (untwisting drives early suction)

E/A reversal ($E/A < 1$ — grade I impaired relaxation)

- ASE/EACVI Diastolic Function Guidelines 2016. J Am Soc Echocardiogr 2016;29(4):277–314.

4.3 Systolic Dysfunction (DCM/HFrEF)

ESV ~84 mL, EF ~30%

- Jin SM et al. J Korean Med Sci 2007;22(4):633–8.

Near-linear ejection (no biphasic distinction)

Global uniform impairment removes the two-phase ejection pattern.

- CMR volumetric literature

Restrictive filling pattern (tall steep E-wave, truncated, minimal diastasis, small A-wave)

Elevated LA pressure drives rapid early pressure equalisation across stiff dilated ventricle.

- Kovacs SJ et al. Am J Physiol Heart Circ Physiol 2004.
- AHA/ACC Heart Failure Guidelines — restrictive cardiomyopathy pattern

$E/A > 2$ (restrictive — opposite of HFpEF)

- Standard HF classification (AHA/ACC)

5. Pressure

5.1 Healthy

LVEDP ~5 mmHg, rapid IVC rise, peak ~120 mmHg

- Wiggers CJ. 1928.
- Kern MJ. The Cardiac Catheterization Handbook. 5th ed., 2011.

IVR fall — rapid exponential (τ ~35 ms)

→ Dong SJ et al. Am J Physiol Heart Circ Physiol 2001;281:H2002–9.

Flat diastole (~1.5 mmHg) — no LVP rise from atrial contraction

Atrial systole raises LA pressure but the LV itself does not contract. The rise seen in the original CSV was identified as an artefact and removed.

→ Standard physiology

dP/dt highest of all three conditions — contractility marker

→ Kern MJ. The Cardiac Catheterization Handbook. 2011.

5.2 Diastolic Dysfunction (HFpEF)

IVC dP/dt preserved — same rate as healthy

→ Wang J et al. Eur Heart J 2008;29(10):1283–9. (contractility preserved in HFpEF)

Peak systolic ~122 mmHg (same as healthy)

→ Wang J et al. Eur Heart J 2008;29(10):1283–9.

IVR fall slower (prolonged tau — impaired relaxation)

→ Dong SJ et al. Am J Physiol Heart Circ Physiol 2001;281:H2002–9.

→ Takeuchi M et al. Eur Heart J 2007;28(19):2313–21.

Elevated diastolic baseline ~10–12 mmHg (elevated LVEDP)

→ ESC Heart Failure Guidelines 2021. Eur Heart J 2021;42(36):3599–726.

5.3 Systolic Dysfunction (DCM/HFrEF)

Blunted IVC dP/dt (slow gradual rise)

→ Wang J et al. Eur Heart J 2008;29(10):1283–9.

→ Jin SM et al. J Korean Med Sci 2007;22(4):633–8.

Lower rounded peak ~96 mmHg

→ Wang J et al. Eur Heart J 2008;29(10):1283–9.

IVR fall rate near-normal (relaxation less primarily impaired in DCM vs HFpEF)

→ DCM pathophysiology — systolic impairment predominates

Diastolic baseline ~3 mmHg flat — no atrial kick artefact

Device/mock loop operating baseline.

6. Key Discriminating Features Summary

Feature	Healthy	HFpEF	DCM/HFrEF
Peak twist	~15.5°	~15.5° (preserved)	~4.5° (abolished)
IVR untwisting	40–50% recoil	<10% recoil	Near zero
Untwisting location	IVR	Rapid filling	Throughout (slow)
MAPSE	~13 mm	~11 mm	<7 mm
PSS feature	Subtle	Prominent	Absent
EF	~63%	~57%	~30%
E/A pattern	E-dominant >1	A-dominant <1	Restrictive >2
IVC dP/dt	Highest	Same as healthy	Blunted
LVEDP	~5 mmHg	~12 mmHg	~3 mmHg
Ejection shape	Biphasic	Near-identical to H	Linear
Height recovery	Brisk IVR snap	Slow IVR, delayed	Biphasic hesitation

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- Yip GWK, Zhang Y, Tan PY et al. Left ventricular long-axis changes in early diastole and systole: impact of systolic function on diastole. *Clin Sci (Lond)* 2002;102(5):515–22. doi:10.1042/cs1020515
- Steverson D, Mele JM, McNeil CR, Wells M, Kendall JL. Mitral annular plane systolic excursion as a screening tool for reduced ejection fraction. *J Emerg Med* 2023.
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